T22(iii): statistical concentration of the composite start

Under EXP′ and NL_η (Decision Point 2)

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This document proves part (iii) of T22 only, with the hypotheses and conventions of `STATEMENT.md`. EXP′ is an explicit hypothesis, replacing EXP as approved in Decision Point 2. The only previously proved input is T22(i), quoted in Section 6. No target-posterior consistency, clustering-separation theorem, or other unproved v3 conjecture is used.

1 Setting and conclusion

The centered raw response is $Y = Cy^r = \mu + C\varepsilon$, where $C = I - \mathbf{1}\mathbf{1}^\top/n$, $\varepsilon \sim N_n(0, \sigma^2 I)$, $\sigma^2 > 0$ is fixed, and $\mu = X\theta^*$ is centered and deterministic. The standardized design has centered columns of squared norm $n-1$. The true support S_0 has fixed size s_0; $p \le n^A$, $A \ge 1$; $\|\mu\|^2 \le Mn$; and S_0 is eventually feasible for $K = \min(p, \lfloor n/2 \rfloor)$. All within-block Gram correlation matrices have smallest eigenvalue at least a fixed $\phi_0^2 > 0$. The actual blocks C_b are deterministic functions of X and have sizes at most $B \le 16$. Let $U_b^* = S_0 \cap C_b$.

Write P_T for projection onto X_T and, for $l \in C_b \setminus T$,

$$u_{T,l} = (I - P_T)x_l / \|(I - P_T)x_l\|, \qquad \delta_{T,l} = u_{T,l}^\top \mu, \qquad Z_{T,l} = u_{T,l}^\top \varepsilon/\sigma,$$
$$\Delta_T = \|(I - P_T)\mu\|^2/(n-1), \qquad v_T = \sigma^2 + \Delta_T, \qquad \mathrm{RSS}_T = \|(C - P_T)Y\|^2.$$

The directions are well defined by within-block SE. Since $P_T C = P_T$, $\mathrm{RSS}_T = \|(I - P_T)Y\|^2$ and

$$x_{T,l} := \frac{\mathrm{RSS}_T - \mathrm{RSS}_{T+l}}{\mathrm{RSS}_T} = \frac{(\delta_{T,l} + \sigma Z_{T,l})^2}{\mathrm{RSS}_T}. \tag{1}$$

Here and below $T + l = T \cup \{l\}$. For all sufficiently large n, every within-block residual space has positive dimension, so its Gaussian residual has positive squared norm almost surely; the finitely many $x_{T,l}$ then belong to $[0, 1)$ almost surely. Standardizing the response does not change x.

The exact block tables are $r_b(U) = m(U)/m(\varnothing)$, with the v3 marginal score m defined in the statement. The exact start is

$$\pi_0(S) \;\propto\; w(|S|) \prod_b r_b(S \cap C_b), \qquad |S| \le K, \qquad w(k) = \frac{1}{2\binom{2p-1}{k}}.$$

In particular the tables and start weights are positive and finite. EXP′ assumes, uniformly over *all* within-block additions,

$$\log \frac{r_b(T+l)}{r_b(T)} = -\frac{1}{2}\log(n+1) - \nu \log(1 - x_{T,l}) + R_{n,b,T,l}, \qquad \nu = \frac{n}{2} + 0.01, \tag{2}$$

where $\rho_n = \max_{b,T,l} |R_{n,b,T,l}| = O_{\mathbb{P}_0}(1)$ in the strict version, or $\rho_n = o_{\mathbb{P}_0}(\log n)$ in the weaker version.

Define the deterministic families

$$\mathcal{A}_n = \{(b,T,l) : l \in C_b,\ T \subseteq C_b \setminus \{l\}\}, \qquad \mathcal{L}_n = \{(b,T,l) \in \mathcal{A}_n : l \notin S_0,\ U_b^* \subseteq T\}.$$

On $\mathcal{L}_n$, NL_η assumes $\delta_{T,l}^2 \le \eta\sigma^2 \log n$, with $0 \le \eta < (\sqrt{2A+1} - \sqrt{2A})^2$. Fix $\epsilon_0 > 0$, set $h_n = \sqrt{2(A+\epsilon_0)\log n}$, and put

$$E_{T,l} = \min\left\{ \frac{\delta_{T,l}^2}{2\Delta_T},\ \frac{\delta_{T,l}^2}{2v_T} + \frac{|\delta_{T,l}|\sigma h_n}{v_T} \right\}, \tag{3}$$

with $E_{T,l} = 0$ if $\delta_{T,l} = 0$. If $\Delta_T = 0$ then $\delta_{T,l} = 0$, so this convention covers the zero denominator. The fixed $\kappa > 0$ satisfies the statement's $\mathrm{NL}_{\mathrm{avg}}(\kappa)$:

$$\frac{1}{p - s_0} \sum_{l \notin S_0} \max_{T:\,(b(l),T,l) \in \mathcal{L}_n} \exp(E_{T,l}) \le n^{1/2-\kappa} \qquad (p > s_0), \tag{4}$$

eventually. Section 5 verifies the stated choices supplied by NL_η.

VIS assumes, for some deterministic $M_n^V \to \infty$, that

$$g_{b,U} := \|(P_{U \cup U_b^*} - P_U)\mu\|^2 \ge M_n^V v_U \log n \quad \text{if } U \subseteq C_b,\ U_b^* \not\subseteq U. \tag{5}$$

For $p > s_0$ write

$$u_n = \frac{1}{p - s_0} \sum_{l \notin S_0} \max_{T:\,(b(l),T,l) \in \mathcal{L}_n} \frac{r_{b(l)}(T + l)}{r_{b(l)}(T)},$$

and put $u_n = 0$ if $p = s_0$. The quantities q_n, e_n are exactly the aggregation quantities with reference S_0 in the statement; in particular

$$e_n = 2p \max_{b,U:\,a = |U_b^* \setminus U| > 0} \left\{ \frac{r_b(U)}{r_b(U \cup U_b^*)} \right\}^{1/a}, \tag{6}$$

where the empty maximum is zero. Set $a_n = 2^B \sqrt{\log n}\, n^{-\kappa}$.

Theorem 1 (T22(iii))**.** *Under these hypotheses with the strict EXP′ remainder,*

$$q_n \le u_n = O_{\mathbb{P}_0}(a_n), \qquad e_n = o_{\mathbb{P}_0}(n^{-m}) \text{ for every fixed } m > 0,$$
$$1 - \pi_0(S_0) = O_{\mathbb{P}_0}(a_n + e_n) = O_{\mathbb{P}_0}(a_n).$$

Under the weaker remainder there exist $\xi_n, \zeta_n = o_{\mathbb{P}_0}(1)$ such that, on events with probability tending to one,

$$q_n \le u_n \le a_n n^{\xi_n}, \qquad 1 - \pi_0(S_0) \le a_n n^{\zeta_n} + s_0 e_n.$$

The superpolynomial bound for e_n still holds. In either version, $\pi_0(S_0) \to 1$ in probability. Exact zero leakage allows $\kappa = 1/2$.

2 Uniform residual control

We give the needed Gaussian bounds explicitly. None requires independence between comparisons. Throughout, constants are independent of n, b, T, l. If a family is empty its nonnegative supremum is zero.

Lemma 1. *Let $\mathcal{T}_n = \{T : T \subseteq C_b \text{ for some } b\}$. Then*

$$\sup_{T\in\mathcal{T}_n} |\,\mathrm{RSS}_T - (n-1)v_T| = O_{\mathbb{P}_0}(\sqrt{n\log n}), \qquad \sigma^2 \le v_T \le \sigma^2 + 2M. \tag{7}$$

Consequently $\inf_{T\in\mathcal{T}_n} \mathrm{RSS}_T/n$ *is bounded away from zero with probability tending to one, and*

$$\sup_{T\in\mathcal{T}_n} \left| \frac{\nu}{\mathrm{RSS}_T} - \frac{1}{2v_T} \right| = O_{\mathbb{P}_0}\left(\sqrt{\frac{\log n}{n}} \right). \tag{8}$$

There is a deterministic $C_R < \infty$ for which $\mathbb{P}_0(\|Y\|^2 \le C_R n) \to 1$.

Proof. There are at most $p2^B \le 2^{16} n^A$ members of $\mathcal{T}_n$, and at most this many additions in $\mathcal{A}_n$: indeed $|\mathcal{A}_n| = \sum_b |C_b| 2^{|C_b|-1} \le p2^B$. For every T, put $r_T = (I - P_T)\mu$. Since $P_T \preceq C$,

$$\mathrm{RSS}_T - (n-1)v_T = 2r_T^\top \varepsilon + \{\|C\varepsilon\|^2 - (n-1)\sigma^2\} - \|P_T\varepsilon\|^2. \tag{9}$$

The middle term is $O_{\mathbb{P}_0}(\sqrt{n})$, since $\|C\varepsilon\|^2/\sigma^2$ is chi-squared with $n-1$ degrees of freedom and variance $2(n-1)$. The first term is centered Gaussian with variance at most $4\sigma^2 Mn$. The Gaussian tail bound $\Pr(|N(0,1)| > t) \le 2\exp(-t^2/2)$ and a union bound over $\mathcal{T}_n$ give $\sup_T |r_T^\top \varepsilon| = O_{\mathbb{P}_0}(\sqrt{n\log n})$. If $M = 0$ these terms are zero.

For completeness, for any deterministic orthogonal projection P of rank $d \le 16$,

$$\mathbb{E}_0 \exp\{\|P\varepsilon\|^2/(4\sigma^2)\} = 2^{d/2} \le 2^8.$$

Markov's inequality and a union bound imply

$$\mathbb{P}_0\left(\max_{T\in\mathcal{T}_n} \|P_T\varepsilon\|^2 > L\sigma^2 \log n \right) \le 2^{24} n^{A - L/4} \to 0$$

for any fixed $L > 4A$. This proves (7) from (9). Also $0 \le \Delta_T \le Mn/(n-1) \le 2M$. Dividing (7) by n proves the uniform positive lower bound and, using $\nu = n/2 + 0.01$, proves (8) by taking reciprocals on that event. Finally $\|Y\|^2 \le 2\|\mu\|^2 + 2\|C\varepsilon\|^2$ and $\|C\varepsilon\|^2/n \to \sigma^2$ in probability, so a sufficiently large fixed C_R works. □

Lemma 2 (Quadratic specialization on the leakage family)**.** *On $\mathcal{L}_n$,*

$$\max_{\mathcal{L}_n} x_{T,l} = O_{\mathbb{P}_0}(\log n/n), \qquad \max_{\mathcal{L}_n} n x_{T,l}^2 = o_{\mathbb{P}_0}(1), \tag{10}$$

$$d_n := \max_{\mathcal{L}_n} \left| -\nu \log(1 - x_{T,l}) - \frac{(\delta_{T,l} + \sigma Z_{T,l})^2}{2v_T} \right| = o_{\mathbb{P}_0}(1). \tag{11}$$

In particular the logarithmic expansion is used in quadratic form only on this family, with no extra remainder assumption.

Proof. Each $Z_{T,l}$ is exactly standard normal: $u_{T,l}$ is a deterministic unit vector. Let $\mathcal{H}_n = \{\max_{\mathcal{L}_n} |Z_{T,l}| \le h_n\}$. Counting comparisons as above gives

$$\mathbb{P}_0(\mathcal{H}_n^c) \le 2p2^B \exp(-h_n^2/2) \le 2^{17} n^{-\epsilon_0} \to 0. \tag{12}$$

On $\mathcal{H}_n$, NL_η implies

$$\max_{\mathcal{L}_n} (\delta_{T,l} + \sigma Z_{T,l})^2 \le \sigma^2 \big(\sqrt{\eta} + \sqrt{2(A + \epsilon_0)}\big)^2 \log n.$$

Together with Lemma 1 and (1), this gives (10). For $0 \le x < 1$,

$$0 \le -\log(1-x) - x = \int_0^x \frac{t}{1-t}\,dt \le \frac{x^2}{2(1-x)}.$$

Thus uniformly on $\mathcal{L}_n$, the difference between $-\nu\log(1-x)$ and νx is $O_{\mathbb{P}_0}((\log n)^2/n) = o_{\mathbb{P}_0}(1)$. The difference between νx and the quadratic expression in (11) is, by (8), at most $O_{\mathbb{P}_0}(\log n\sqrt{\log n/n}) = o_{\mathbb{P}_0}(1)$. This proves the claim without conditioning on s_y or treating standardized errors as independent Gaussian errors. □

3 The averaged noise-addition rate

Lemma 3 (A truncated Gaussian integral)**.** *For deterministic $\Delta \ge 0$, $v = \sigma^2 + \Delta$, $\delta \in \mathbb{R}$ (with $\delta = 0$ if $\Delta = 0$), and $Z \sim N(0,1)$,*

$$\mathbb{E}\left[\exp\left\{\frac{(\delta+\sigma Z)^2}{2v}\right\}\mathbf{1}\{|Z| \le h\}\right] \le \frac{2h}{\sqrt{2\pi}}\exp\left[\min\left\{\frac{\delta^2}{2\Delta}, \frac{\delta^2}{2v} + \frac{|\delta|\sigma h}{v}\right\}\right] \tag{13}$$

for $h > 0$, with the minimum interpreted as zero when $\delta = 0$.

Proof. After including the Gaussian density the exponent inside the integral is

$$f(z) = \frac{\delta^2}{2v} + \frac{\delta\sigma}{v}z - \frac{\Delta}{2v}z^2.$$

For $|z| \le h$, dropping the nonpositive last term gives $f(z) \le \delta^2/(2v) + |\delta|\sigma h/v$. If $\Delta > 0$, completing the square gives

$$f(z) = \frac{\delta^2}{2\Delta} - \frac{\Delta}{2v}\left(z - \frac{\delta\sigma}{\Delta}\right)^2 \le \frac{\delta^2}{2\Delta}.$$

If $\delta = 0$, $f(z) \le 0$, also when $\Delta = 0$. Integrating the resulting uniform bound over $[-h, h]$ proves (13). □

Proposition 1. *With $a_n = 2^B\sqrt{\log n}\,n^{-\kappa}$, the strict remainder gives $u_n = O_{\mathbb{P}_0}(a_n)$. Under the weaker remainder, $u_n \le a_n n^{\xi_n}$ on events of probability tending to one for some $\xi_n = o_{\mathbb{P}_0}(1)$, and hence $u_n \to 0$ in probability.*

Proof. If $p = s_0$ the claims are immediate since $u_n = 0$. For indices with $p > s_0$, define the nonnegative random variable

$$V_n = \frac{1}{p - s_0}\sum_{l \notin S_0}\sum_{T:\,(b(l),T,l)\in\mathcal{L}_n}\exp\left\{\frac{(\delta_{T,l} + \sigma Z_{T,l})^2}{2v_T}\right\}\mathbf{1}\{|Z_{T,l}| \le h_n\}.$$

There are at most 2^B terms in the inner sum. Applying Lemma 3 to each term and using (4) yields

$$\mathbb{E}_0 V_n \le \frac{2h_n}{\sqrt{2\pi}}\,2^B n^{1/2-\kappa}. \tag{14}$$

Linearity of expectation is all that is used; the comparisons need not be independent. Markov's inequality therefore gives

$$W_n := \frac{V_n}{2^B\sqrt{\log n}\,n^{1/2-\kappa}} = O_{\mathbb{P}_0}(1). \tag{15}$$

On $\mathcal{H}_n$, use (2), Lemma 2, and the inequality maximum $\le$ sum to obtain

$$u_n \le (n+1)^{-1/2} \exp(\rho_n + d_n) V_n \le a_n \exp(\rho_n + d_n) W_n. \tag{16}$$

No independence between W_n and the remainder is required. If $\rho_n = O_{\mathbb{P}_0}(1)$, then $\exp(\rho_n + d_n) = O_{\mathbb{P}_0}(1)$, so their product is bounded in probability. Since $\mathbb{P}_0(\mathcal{H}_n^c) \to 0$, this proves $u_n = O_{\mathbb{P}_0}(a_n)$.

For the weaker version let

$$\xi_n = \frac{\rho_n + d_n + \log \max(1, W_n)}{\log n}$$

when $p > s_0$, and set $\xi_n = 0$ otherwise. Since $\log \max(1, W_n) = O_{\mathbb{P}_0}(1)$, we have $\xi_n = o_{\mathbb{P}_0}(1)$ and (16) gives the asserted high-probability inequality. Because $B \le 16$ and $\kappa > 0$, on $\{\xi_n \le \kappa/2\}$ its right side is at most $2^{16}\sqrt{\log n}\, n^{-\kappa/2} \to 0$. The same concentration follows directly in the strict version. These arguments also cover sequences alternating between $p = s_0$ and $p > s_0$. □

4 Visibility and the completion penalty

Proposition 2. *Under either remainder condition, $e_n = o_{\mathbb{P}_0}(n^{-m})$ for every fixed $m > 0$.*

Proof. If the index family in (6) is empty, $e_n = 0$. Otherwise for each pair (b, U) in that family put $V = U \cup U_b^*$, $a = |U_b^* \setminus U|$, and $P = P_V - P_U$. Nestedness of the subspaces and SE show that P is a deterministic orthogonal projection of rank a, with $1 \le a \le B \le 16$. There are at most $p2^B$ pairs. The same exponential-moment and union-bound argument used in Lemma 1 gives

$$\max_{b,U} \|P\varepsilon\| = O_{\mathbb{P}_0}(\sqrt{\log n}). \tag{17}$$

Since $P \preceq C$, $PY = P\mu + P\varepsilon$. By VIS, $\|P\mu\| = \sqrt{g_{b,U}} \ge \sigma\sqrt{M_n^V \log n}$. It follows from (17) and $M_n^V \to \infty$ that, simultaneously for every pair with probability tending to one,

$$\mathrm{RSS}_U - \mathrm{RSS}_V = \|PY\|^2 \ge \left(\|P\mu\| - \|P\varepsilon\|\right)^2 \ge \frac{1}{4} g_{b,U}. \tag{18}$$

Here the middle inequality is used on the event $\|P\varepsilon\| \le \|P\mu\|/2$ for every pair; its probability tends to one by the preceding bounds.

List the a missing true variables in any fixed order and let $U = T_0 \subset T_1 \subset \cdots \subset T_a = V$ be the resulting within-block addition path. Summing EXP′ along this path is exact apart from the stated remainders:

$$\log \frac{r_b(V)}{r_b(U)} \ge -\frac{a}{2} \log(n+1) + \nu \log \frac{\mathrm{RSS}_U}{\mathrm{RSS}_V} - a\rho_n. \tag{19}$$

Indeed $1 - x_{T_{i-1}, l_i} = \mathrm{RSS}_{T_i} / \mathrm{RSS}_{T_{i-1}}$, so the logarithms telescope. No quadratic approximation is made on this path. For $0 < t \le 1$, $-\log t \ge 1 - t$. On the simultaneous event from (18) and $\|Y\|^2 \le C_R n$, since $\mathrm{RSS}_U \le \|Y\|^2$ and $\nu \ge n/2$,

$$\nu \log \frac{\mathrm{RSS}_U}{\mathrm{RSS}_V} \ge \nu \frac{\mathrm{RSS}_U - \mathrm{RSS}_V}{\mathrm{RSS}_U} \ge \frac{g_{b,U}}{8C_R} \ge c M_n^V \log n, \qquad c := \frac{\sigma^2}{8C_R} > 0. \tag{20}$$

Combining (19)–(20), taking ath roots of the reciprocal ratio, and using $a \le 16$, gives

$$\left\{\frac{r_b(U)}{r_b(V)}\right\}^{1/a} \le \exp\left\{\frac{1}{2}\log(n+1) + \rho_n - \frac{c}{16} M_n^V \log n\right\}.$$

Thus, on an event of probability tending to one,

$$e_n \le 2n^A (n+1)^{1/2} \exp(\rho_n) n^{-cM_n^V/16}. \tag{21}$$

In either remainder regime $\rho_n / \log n = o_{\mathbb{P}_0}(1)$. For fixed $m > 0$, the logarithm of n^m times the right side, divided by $\log n$, is

$$m + A + \tfrac{1}{2} - \frac{c}{16} M_n^V + o_{\mathbb{P}_0}(1) + o(1) \;\longrightarrow\; -\infty$$

in probability. Therefore $n^m e_n \to 0$ in probability, including the negligible complement of the displayed event. This proves the result. □

5 Admissible rates supplied by NL_η

For every leakage comparison, the second branch of (3) and $v_T \ge \sigma^2$ imply

$$E_{T,l} \le \left\{ \frac{\eta}{2} + \sqrt{2\eta(A+\epsilon_0)} \right\} \log n. \tag{22}$$

Hence (4) follows whenever

$$0 < \kappa \le \frac{1}{2} - \frac{\eta}{2} - \sqrt{2\eta(A+\epsilon_0)}.$$

To check that positive choices exist, put $t = \sqrt{\eta}$. At $\epsilon_0 = 0$ the condition for a positive upper endpoint is $t^2 + 2\sqrt{2A}\, t < 1$, equivalent for $t \ge 0$ to $t < \sqrt{2A+1} - \sqrt{2A}$. This is exactly the assumed strict threshold. Continuity allows a sufficiently small fixed $\epsilon_0 > 0$. When $\eta > 0$, every fixed $0 < \kappa < 1/2 - \eta/2 - \sqrt{2A\eta}$ is allowed in this way. When $\eta = 0$, all leakage $\delta_{T,l}$ vanish, so $E_{T,l} = 0$; the average is 1 if nonempty and $\kappa = 1/2$ is allowed. Neither this calculation nor NL_η replaces EXP′.

6 Aggregation and completion of the proof

We use the previously proved T22(i), with feasible reference S_0:

$$q_n \le u_n, \qquad 1 - \pi_0(S_0) \le \frac{(s_0+1)u_n}{1-u_n} + s_0 e_n \quad \text{on } \{u_n < 1\}. \tag{23}$$

This input holds for every partition with positive tables, with the cap included; the actual blocks satisfy $|C_b| \le B \le K$, since $\lfloor \sqrt{n} \rfloor \le \lfloor n/2 \rfloor$ for $n \ge 3$ and $\lfloor \sqrt{p} \rfloor \le p$, so the wave-1 hypotheses hold verbatim. Its source is the wave-1 T22 proof and the T22 chapter of `COLLECTED_PROOFS_v2.tex`. The bound is also quoted verbatim in `STATEMENT.md`, Section 5. It does not rely on part (iii).

Proof of Theorem 1. By Proposition 1, $u_n \to 0$ in probability in both remainder regimes, so $\mathbb{P}_0(u_n \le 1/2) \to 1$. On that event,

$$1 - \pi_0(S_0) \le 2(s_0+1)u_n + s_0 e_n. \tag{24}$$

The strict version follows from $u_n = O_{\mathbb{P}_0}(a_n)$ and Proposition 2. More explicitly, division of (24) by $a_n + e_n > 0$ bounds the ratio by $2(s_0+1)u_n/a_n + s_0$, which is bounded in probability; the event where (24) need not hold has probability tending to zero. Choosing any $m > \kappa$ in Proposition 2 gives $e_n/a_n \to 0$ in probability, proving the sharper $O_{\mathbb{P}_0}(a_n)$ form.

For the weaker version, intersect $\{u_n \le 1/2\}$ with the event in Proposition 1 and set $\zeta_n = \xi_n + \log(2(s_0+1))/\log n = o_{\mathbb{P}_0}(1)$. Equation (24) then gives the asserted exponent-form bound. It tends to zero in probability, by the same $\kappa/2$ argument as in Proposition 1, and because $e_n \to 0$. The bound $q_n \le u_n$ is deterministic by (23).

All empty-family conventions are respected: when $p = s_0$, $q_n = u_n = 0$ and (23) bounds the error by $s_0 e_n$; when $s_0 = 0$, $e_n = 0$ and VIS is vacuous. Section 5 proves the final claim about exact zero leakage. □

Remark 1 (Scope and interpretation). The conclusion concerns the probability law of a single exact start draw, through its mass $\pi_0(S_0)$. It asserts neither that all initial particles equal S_0 nor schedule completion, evidence accuracy, target concentration, or any property of cut groups. No independence between block comparisons, no numerical experiment, and no unproved design-prevalence claim enters the argument. The logarithmic EXP$'$ assumption is retained throughout; its quadratic specialization is justified only where NL_η applies.